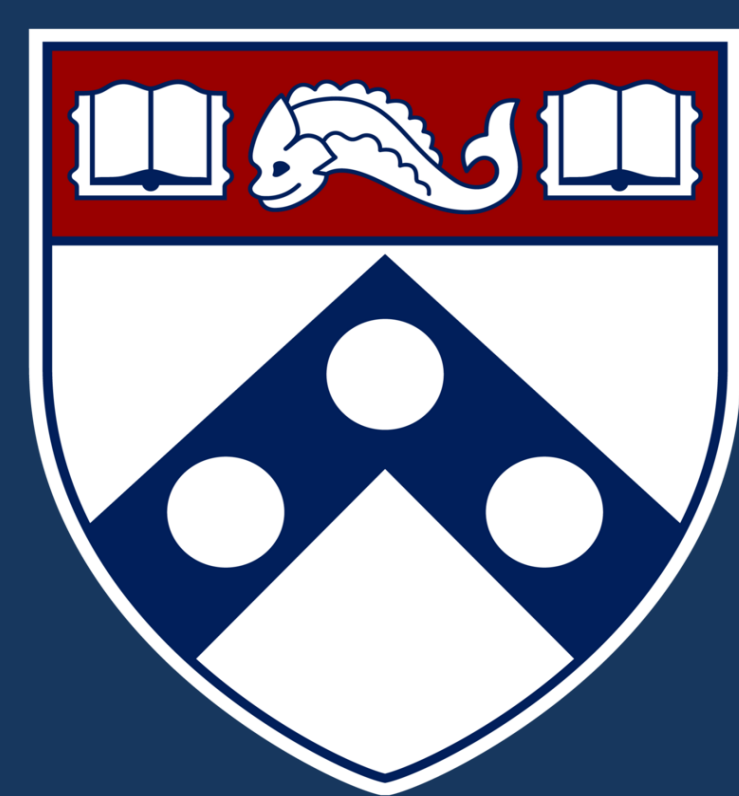


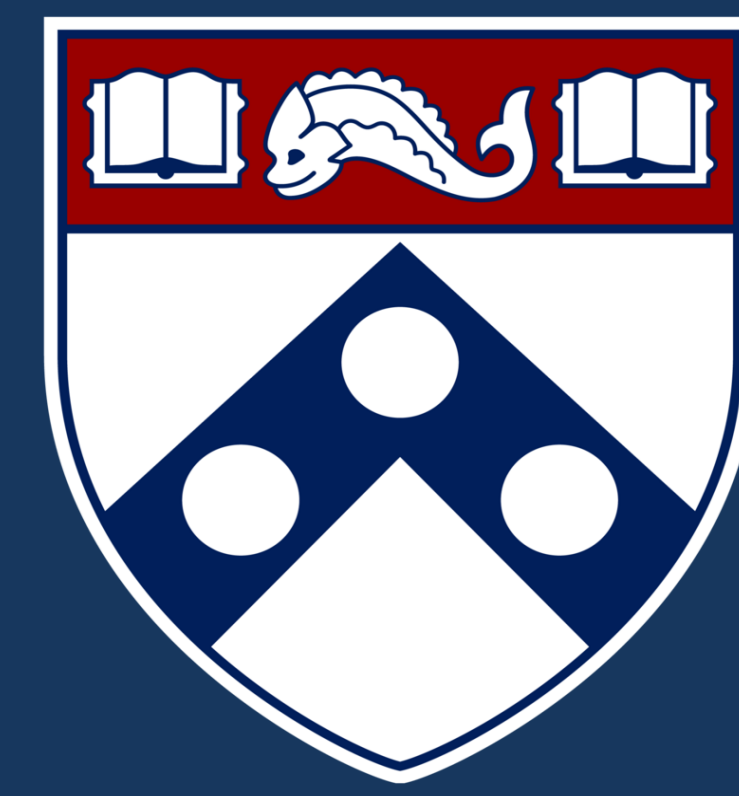


Robust Compute-in-Memory for Edge Healthcare

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1. MOTIVATION

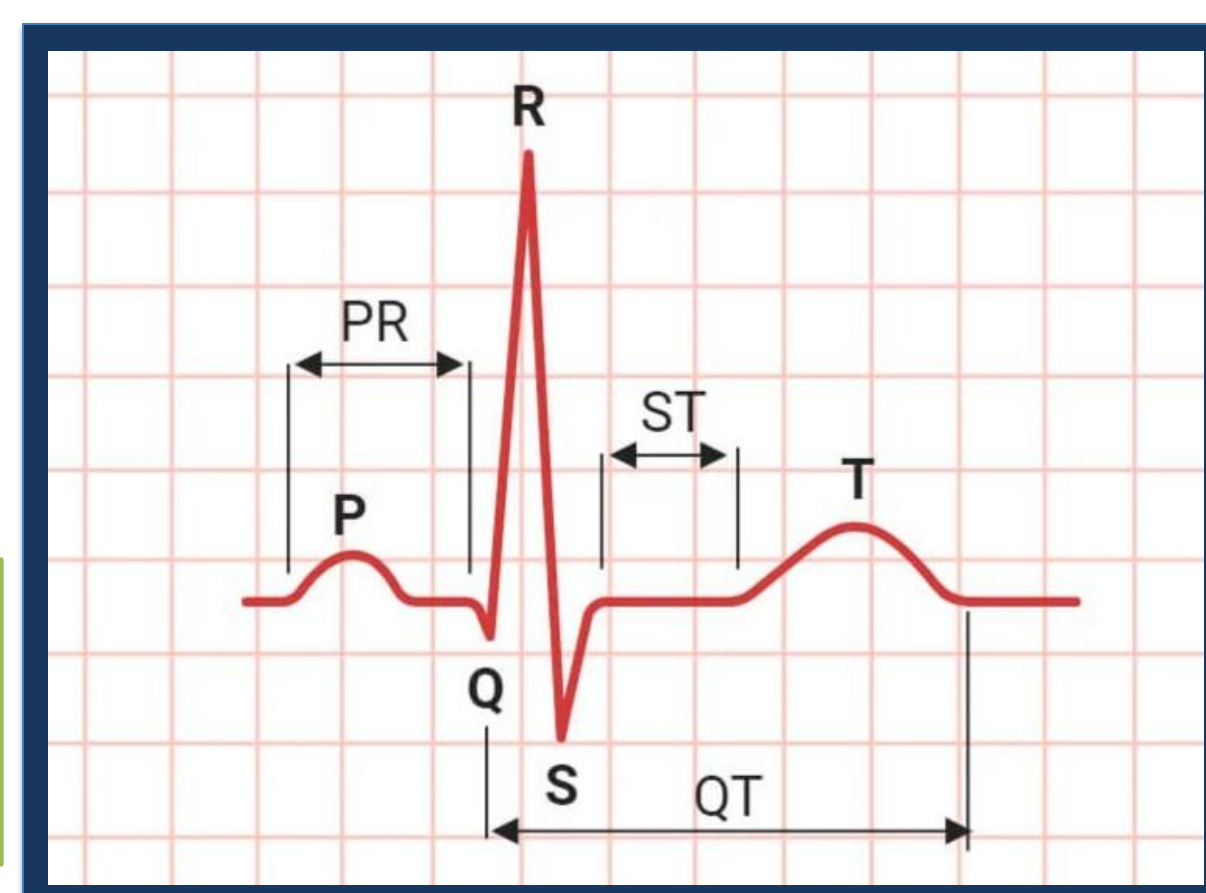
THE PROBLEM

1000X

RF tx energy vs local inference

60%

AI energy from data movement



Wearable ECG patches stream raw data over BT/Wi-Fi, which is unsustainable for battery life.

Von Neumann wall: DRAM↔ALU data shuffle dominates latency and energy.

SOLUTION: ANALOG COMPUTE-IN-MEMORY

Perform multiply-accumulate (MAC) operations directly inside a non-volatile memory crossbar. No data movement, massively parallel, analog.

$$I_{out} = \sum G_{ij} \cdot V_i$$

THE HARDWARE CHALLENGE

OxRAM

Device Parameters

R_ON	1–10 kΩ
R_OFF	100 kΩ–1 MΩ
V_READ	0.2 V
CELL	2T2R differential

Top Electrode

HfO₂
(oxide)

Bottom Electrode

Failure mode: Random Telegraph Noise (RTN)

Model: $G = G \cdot \exp(N(0, \sigma^2))$ with $\sigma = 0.10$

PCM

Device Parameters

R_ON	10–100 kΩ
R_OFF	1–10 MΩ
V_READ	0.3 V
MATERIAL	GST (amorph./cryst.)

Top Electrode

Phase Change Material

Isolator Heater Isolator

Bottom Electrode

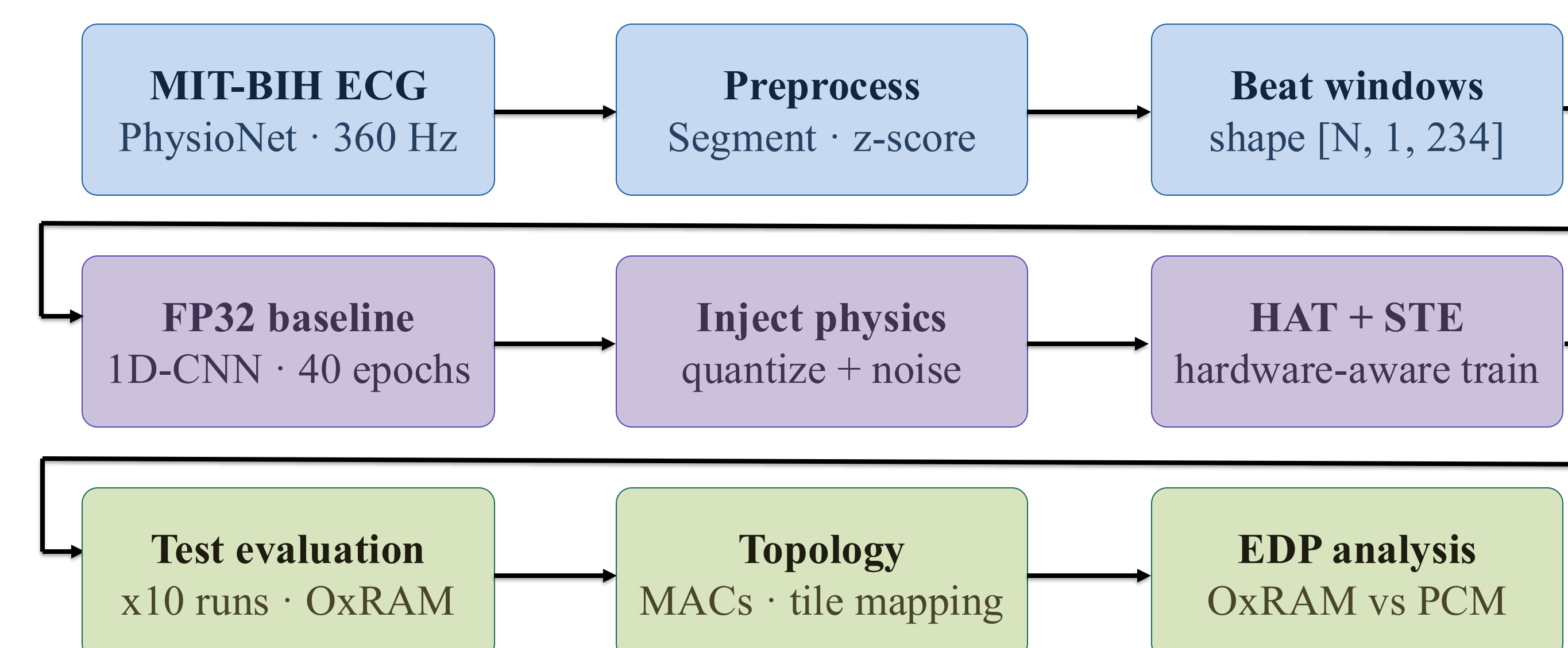
Failure mode: Resistance Drift (Structural Relaxation)

Model: $R(t) = R_0 \cdot (t/t_0)^{-\nu}$ with $\nu = 0.05$

OBJECTIVES

- Compare OxRAM vs PCM accuracy degradation on ECG arrhythmia classification.
- Implement HAT + STE to recover accuracy under hardware noise.
- Quantify EDP trade-offs between OxRAM and PCM.

2. METHODOLOGY



DATASET: MIT-BIH ARRHYTHMIA DATABASE

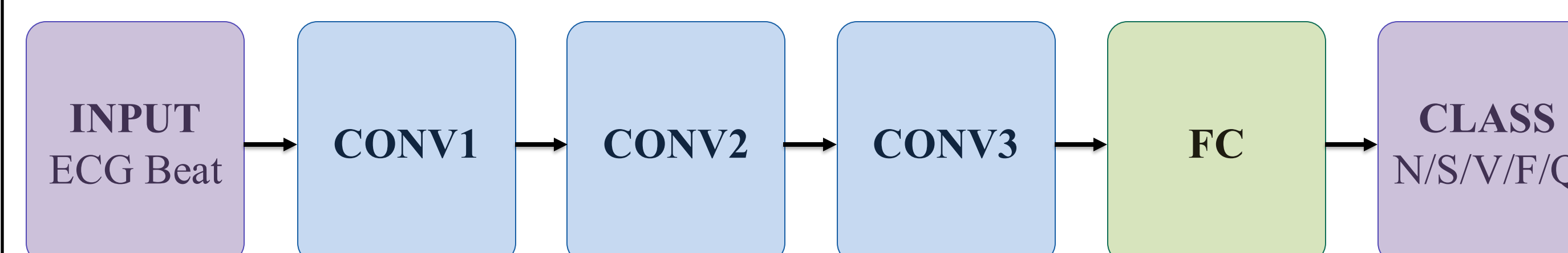
Source: PhysioNet, 48 half-hour recordings, 360 Hz, 2-lead ECG.

Split: De Chazal inter-patient protocol - 22 train / 5 val / 20 test recordings. No patient appears in two splits.

Labels: 5 class - N (Normal), S (Supraventricular), V (Ventricular), F (Fusion), Q (Unknown).

Window: 90 samples pre-R-peak + 144 post = 234 samples per beat (asymmetric to capture full PQRST).

TinyECGCNN ARCHITECTURE



TOTAL: 682,400 MACs per heartbeat inference.

HARDWARE-AWARE TRAINING (HAT) - FORWARD PASS

1 Quantize (4-bit STE)

Snap FP32 weights to 16 discrete conductance levels. Custom torch.autograd.Function: forward = round-and-rescale; backward = identity (Straight-Through Estimator) so gradients flow past the non-differentiable round.

2 Differential split (2T2R cell)

$G^+ = \text{clamp}(W, 0)$ and $G^- = \text{clamp}(-W, 0)$. Two physical conductances per signed weight and recovers $W = G^+ - G^-$ at the sense amplifier.

3 Inject noise

OxRAM RTN: $G_{\text{noisy}} = G \cdot \exp(N(0, \sigma^2))$, $\sigma=0.10$ - stochastic realization.
PCM Drift: $G_{\text{noisy}} = G \cdot (t/t_0)^{-\nu}$, $\nu=0.05$ - deterministic conductance decay.

4 Recombine and convolve

$W_{\text{eff}} = G^+_{\text{noisy}} - G^-_{\text{noisy}} \rightarrow F.\text{conv1d}(\text{input}, W_{\text{eff}})$. The optimizer sees noisy weights at every batch and adapts toward flat-minima configurations robust to analog perturbation.

3. RESULTS AND CONCLUSION

ML PERFORMANCE - TEST SET EVALUATION

Technology	Accuracy	Macro-F1	F1 (N)	F1 (S)	F1 (V)
FP32 Baseline	65.06%	0.330	0.803	0.194	0.649
HAT-OxRAM	54.67% ± 1.15%	0.320	0.694	0.235	0.623
HAT-PCM	68.15%	0.324	0.846	0.110	0.662

OxRAM: 10% accuracy drop under active RTN, but Macro-F1 falls only 0.010. S-class detection improves.

PCM: 68.2% accuracy - uniform 0.708× drift absorbed by BatchNorm's learnable scale parameters.

SYSTEM-LEVEL EDP ANALYSIS

OxRAM

Total Energy

41.17 nJ

Latency

4.35 μs

Area

0.0419 mm²

Energy-Delay Product

1.79×10^{-13} J·s

PCM

Total Energy

27.52 nJ

Latency

5.82 μs

Area

0.0419 mm²

Energy-Delay Product

1.60×10^{-13} J·s

KEY FINDINGS

- HAT is effective
- Model fits 3 crossbar tiles
- PCM wind on EDP (Energy-Delay-Product)

LIMITATIONS

- PCM drift modeled as a uniform global scalar, state-dependent $v(G)$ not yet captured.
- F and Q classes near-zero F1 due to severe class imbalance in de Chazal split (8 Q training samples).

FUTURE WORK

- State-dependent PCM drift model: $v(G) \propto$ programmed resistance level
- SMOTE oversampling for F and Q minority class recovery
- Multi-bit weight precision sweep (2-bit to 8-bit)